Vireen Chowdary Vesangi

Business Analyst

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Summary

Business analyst combining an MBA in Information Systems & Analytics with a B.Tech in Artificial Intelligence Engineering — able to gather requirements, analyze processes, and back every recommendation with hands-on data work in SQL, Python, Power BI, Tableau, and Excel. Internship experience delivering leadership-facing financial and operations analysis, plus a portfolio of independently built decision-support tools. Comfortable translating between technical and business teams, structuring ambiguous problems into decision-ready frameworks, and presenting insights that change what stakeholders do next.

Skills

Business Analysis: Requirement gathering (BRD/FRD), stakeholder communication & management, process & workflow analysis, gap analysis, KPI design & tracking, decision support, business reporting, management presentations, cross-functional collaboration, project ownership.

Data & Query: SQL (joins, aggregations, subqueries, query optimization), Python (Pandas, NumPy, scikit-learn), Advanced Excel (Pivot Tables, VLOOKUP, formulas, VBA), R (basics).

BI & Visualization: Power BI (DAX, Power Query, Data Modelling, Row-Level Security), Tableau (LOD expressions, parameters, interactive dashboards), Streamlit, data storytelling.

Statistics & Analysis: Hypothesis testing, regression, ANOVA, MANOVA, Chi-Square, Mann-Whitney U, Spearman correlation, K-Means segmentation, forecasting (Prophet), Monte Carlo simulation; SPSS, PSPP.

AI & Emerging Tech: Local LLM workflows (Ollama, Pydantic), BERT-based NLP, human-in-the-loop AI copilot design, GenAI integration into BI workflows.

Domain Exposure: Financial & receivables analytics, retail strategy (Cesim simulation), operations scale-up frameworks, consumer research, healthcare analytics.

Experience

Business Analyst Intern — **May – Jul 2025**

Symbiosys Technologies, Visakhapatnam

- Worked directly under the VP of Operations analyzing 1,500+ multi-currency invoices for an international IT-services firm; owned the workflow end-to-end from raw exports to leadership recommendations.
- Gathered reporting requirements from finance and operations stakeholders and translated them into three interactive Tableau dashboards: revenue & services performance, overdue payments, and monthly/quarterly trends against targets.
- Cleaned and joined client and invoice datasets using SQL and Advanced Excel — resolving date-format inconsistencies and field errors; built a Python currency converter (real-time forex API with fallback) to standardize INR/JPY/SGD/AUD invoices.
- Quantified ~Rs 2.7 Cr in total overdue exposure with ~Rs 1.2 Cr concentrated in a single overseas market and one client holding ~Rs 72 L in receivables; identified the Engineering Consultancy vertical as driving ~27% of revenue.
- Ran Spearman correlation in SPSS (ρ = -0.001, p = 0.989) proving invoice size did **not** predict payment delay — shifting management focus from invoice value to client-behaviour patterns; K-Means clustering (3 segments) revealed elevated delinquency on 45-day terms, leading to credit-policy recommendations presented in management reviews.

Artificial Intelligence Intern — **Feb – May 2023**

Ozonetel Communications Pvt. Ltd., Hyderabad

- Translated business use cases into technical execution on Agent Assist, a production NLP product for contact-centre operations — an early human-in-the-loop AI workflow with the model drafting and human agents acting.
- Built the customer-facing front-end connecting BERT/LLM-driven backends to a usable interface, and supported model development and validation on unstructured customer-interaction data.

Featured Project

Pilot-to-Scale Decision Support for Autonomous Robotics — *Power BI, Python, Streamlit, Ollama*

- Built an end-to-end business analysis framework for autonomous-robotics organizations moving from pilot deployments to large-scale operations, structured around a 5-stage Pilot-to-Scale Lifecycle with explicit go/hold/stop decision logic per pilot.
- Designed a Pilot Readiness Dashboard in Power BI on synthetic operational data (5 fact/dimension tables, 4 global pilots) with DAX measures, weighted readiness scoring, and fleet-level rollups.
- Prototyped a human-in-the-loop "Robotics BA Copilot" (Python, Streamlit, Ollama, Pydantic-validated schemas) converting unstructured operational text — pilot notes, incident reports, customization requests — into structured BA artifacts; privacy-first, fully local inference.
- Packaged as a portfolio artifact (LaTeX report, dashboard, copilot, video walkthrough) and used in outreach to autonomous-robotics companies; led to a product-manager-track interview conversation.

Portfolio Projects

Captain's Call — Probabilistic IPL Match Simulator (Solo, Deployed) *Python, FastAPI, React*

- Built and shipped a live, data-driven T20 match simulator end-to-end: pandas ETL over 1,244 matches / 295,733 deliveries (18 seasons), a probabilistic outcome model with empirical-Bayes shrinkage, a Monte Carlo win-probability engine (2,000 simulations per match), FastAPI backend, React frontend, cloud deployment.
- Demonstrates the analyst habit that matters most: when evaluation data contradicted the design intuition (a 64/36 bias in win probability), diagnosed the root cause, fixed it, and verified the fix empirically.

GeoSpot — Multi-Signal Image Country Prediction *PyTorch, OpenCLIP, YOLOv8*

- Rebuilt a 2022 university project into a modern system fusing scene, text, and flag signals with a trust-weighted fusion layer; setting fusion weights from measured precision (not intuition) improved top-1 accuracy from 66.7% to 73.9%.

Academic Projects

Airbnb Pricing Analysis — Cross-City European Market *Excel, PSPP, Tableau, Power BI*

- Analyzed a multi-city European Airbnb dataset (Amsterdam, Paris, Rome, Vienna, and more) covering weekday/weekend prices, room types, host status, and proximity metrics.
- Built a Linear Regression pricing model (bedrooms, person capacity, attraction index, distance to centre); validated significant price differences across room types with ANOVA (Entire home Rs 447 > Private Rs 255 > Shared Rs 199) and tested room-type vs superhost independence with Chi-Square.
- Segmented listings into 5 K-Means clusters (Budget to Ultra-Luxury; dominant budget cluster of 3,948 listings) and presented findings via Tableau and Power BI dashboards.

AR/VR Customer Experience — Business Research Methods *SPSS, Python (Colab), NLP*

- Designed a mixed-methods study (cross-sectional survey + retailer interviews) grounded in the Technology Acceptance Model and Schwartz's Paradox of Choice; collected 134 consumer responses via stratified sampling.
- Tested 4 hypotheses: MANOVA (Wilks' Lambda $p = 0.001$) on AR/VR usage vs brand loyalty/trust/privacy; multiple regression (Brand Loyalty model $R^2 = 0.448$, Trust strongest predictor); Mann-Whitney U (chosen over t-test due to mixed Likert scales); and one-way MANOVA ($p = 0.035$) confirming higher customization raised satisfaction **and** anxiety/fatigue — the paradox-of-choice effect.
- Ran Python NLP keyword-frequency analysis on retailer interview transcripts, surfacing adoption barriers: cost, staff training, and perceived complexity.

Retail Sales & Returns Analytics — 6-Dashboard Tableau Workbook *Tableau*

- Built a six-dashboard interactive workbook on the Sample Superstore dataset: time-trend analysis (2014–2017), category/sub-category performance, US geographic distribution, customer segment & shipping insights, and a two-part returns & discount impact analysis identifying products at risk.
- Used a wide chart vocabulary — tree maps, bullet charts, box-and-whisker, heat maps, bubble scatters, stacked bars, geographic maps — with LOD expressions and parameters.

Community Health Analytics — 5-Dashboard Power BI Workbook *Power BI, DAX*

- Built dashboards covering population health, patient risk stratification (top 10% high-risk patients by composite risk score), chronic disease monitoring (HbA1c, adherence vs complication risk), social determinants of health, and public-health surveillance across 17+ Indian state clusters.

Cesim Retail Simulation — MBA Phygital Lab *Business strategy, pricing analytics*

- Competed across six rounds of a multi-region retail simulation with team decisions on product pricing, placement, channel mix, marketing spend, and demand forecasting; analyzed round-by-round performance to iteratively optimize revenue and profitability.

Education

MBA — Information Systems, Analytics & International Business **2024 – 2026**
GITAM University, Visakhapatnam *CGPA: 7.6/10*

B.Tech — Artificial Intelligence Engineering (Robotics Specialization) **2019 – 2023**
Amrita Vishwa Vidyapeetham, Amritapuri *CGPA: 7.5/10*

Certifications

- Microsoft PL-300 Power BI Data Analyst (in progress)
- Microsoft Power BI Data Analyst (Coursera)
- Microsoft Engage 2021
- Shore Fest Datathon 1st Place Finish